

Intro to Inverse Kinematics

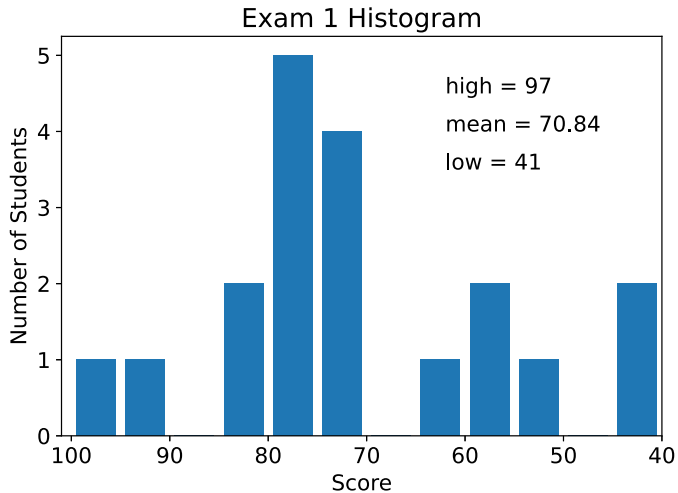
EGR 445/545

GVSU

Plan for Today

- exam 1 solution
- intro to inverse kinematics
- project time

Exam 1 Histogram



Intro to Inverse Kinematics

What do we mean by inverse kinematics?

- what is the definition of kinematics?
- what do we mean by inverse in this case?

solving for joint angles to
reach desired tip position
and orientation

↳ what direction is the
final link/tip pointing?

Inverse Kinematics is Hard

- why?
- why can't I just make it easy?
- what did we have to do to find θ_1 and θ_2 for project 1?
 - was there an easier way?

Complaint of a former student

- “Stop talking so much about what makes a valid problem statement and give me a repeatable way to solve inverse kinematics problems.”
 - what are they really asking for?
 - why can't I meet this request?
 - why do *you* need to know what makes a valid inverse kinematics problem statement?

You can't solve an unsolvable problem

- your customer *probably* doesn't have better technical skills than you
- your boss *probably* doesn't have better technical skills than you
- if someone asks you to do something that cannot be done, what are you going to do?

Possible causes of unsolvable problems

- reachable workspace
- dextrous workspace
- # DOFs vs. # of joints

all points the tip can reach

all points the tip can reach while also being able to achieve all orientation

An Inverse Kinematics Process: The Best I Can Do

- can you locate the wrist?
 - do you have enough information?
- what can you learn from a top view? ←
- what can you learn from a side view? ←
 - what frame is the side view in?

$\hat{x}_1 / \hat{z}_1$ plane

Industrial Robot Inverse Kinematics

- how many joints do most industrial robots have?
 - why?
- what would a valid problem statement for inverse kinematics need to include?
 - how would we solve such a problem?

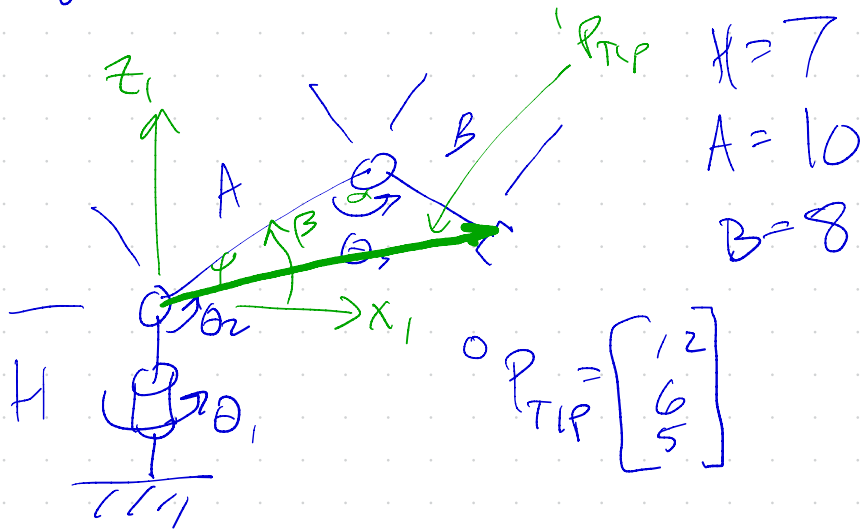
What makes a valid inverse kinematics solution?

- find all solutions
- do not use numeric search algorithms
 - why not?
- find explicit solutions
- only trust `atan2`

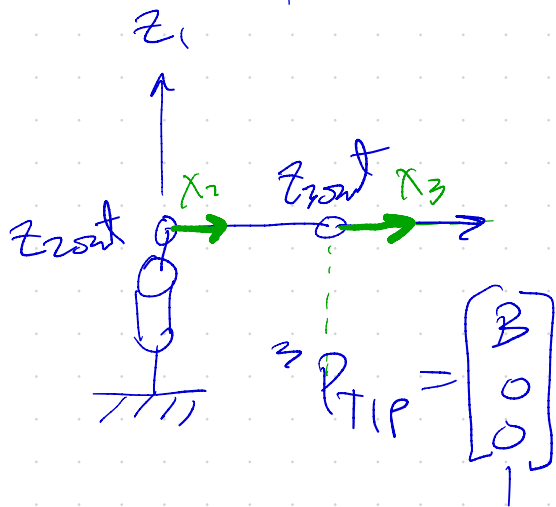
Geometric vs. Algebraic Solution

- but I'm awesome at algebra. . .
 - can you find an algebraic solution for project 1?
- see section 4.4

Project 2 Inv. Kinematics

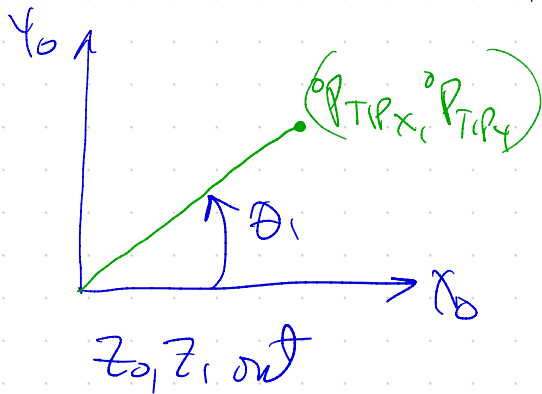


Home position



i	α	a	θ	d
1	0	0	θ_1	H
2	90	0	θ_2	0
3	0	A	θ_3	0

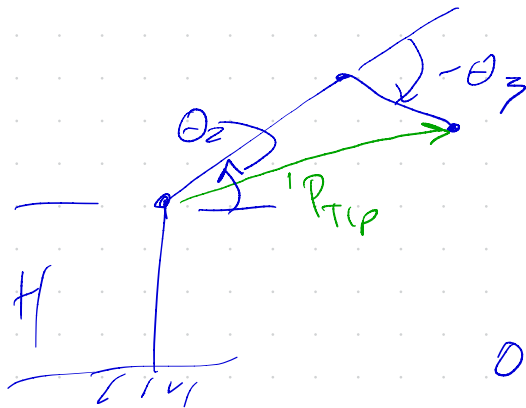
Top View



$$\theta_1 = \arctan 2(p_{TIRy}, p_{TIRx})$$

~~std~~

Side View



$${}^1P_{TIP} = {}^1_0T \cdot {}^0P_{TIP}$$



$${}^1_0T = {}^0_1T^{-1}$$

$${}^0_1T = DH(0, 0, \theta, H)$$

→ solve for θ_2 and θ_3 from
side view two link inv.

→ verify solution using forward
kinematics

$${}^0P_{TIP_check} = {}^0T_1 \cdot {}^1T_2 \cdot {}^2T_3 \cdot {}^3P_{TR}$$

Intro Example

- consider a simplified project 2 robot